EXPLORATORY DATA ANALYSIS OF THE TITANIC DATASET

1. OBJECTIVE

The objective of this project is to explore the Titanic dataset using Python and identify patterns, trends, missing values, and relationships between variables.

2. TOOLS USED

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab

3. KEY FINDINGS

- The dataset contains more passengers who did not survive than passengers who survived.
- Female passengers had a higher survival rate than male passengers.
- Passenger class was associated with survival outcomes. First-class passengers generally had better survival outcomes than third-class passengers.
- The Age column contains missing values, while the Cabin column has a large amount of missing information.
- Ticket fares are right-skewed, meaning most passengers paid lower fares, while a small number paid unusually high fares.
- Correlation analysis and pairplots helped identify relationships and patterns among numerical variables.

4. CONCLUSION

Exploratory Data Analysis helped identify important patterns in passenger survival, gender, passenger class, age, and ticket fares. Statistical summaries and visualizations made the dataset easier to understand. The analysis also highlighted missing values and unusual observations that may require further investigation.